

Horatio Montero Hamkins

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Interdisciplinary engineer with research experience in soft robotics, human computer interaction, industrial engineering, and medical physics

Education

MEng(Hons) In Mechanical Engineering | University of Bath

2021 - 2026

- **Master's Thesis** Development of CNC Wire Bending Machine and Software for Real-Time Portable Art Fabrication (Grade: First-Class Honours), supervised by Dr. Evros Loukaides.
- Awarded the **Arthur Clements Award** for 'the promotion and encouragement of the Art of Mechanical Engineering' and shortlisted for the university-wide **Student Education Award** for 'Outstanding Contribution to Education at Bath'.

Publications

Horatio Hamkins, Ka Ho Tam, Angela Sobremonte, Sandeep Jogi, Eugene Koay, Comron Hassanzadeh, Paul Segars, Neelam Tyagi, Ergys Subashi (2026). Automated Design of Patient-Specific 4D-Printed Phantoms for Quality Assurance of Adaptive Radiotherapy on a 1.5T MR-Linac. *3D Printing in Medicine* (currently under review) [\[link\]](#)

Horatio Hamkins, Christelle Grandvallet, Frédéric Vignat (2026). Extended Reality-Based Human-Robotic Collaborative Welding Systems: A Review. *The International Journal of Advanced Manufacturing Technology* [\[link\]](#)

Lukas Rambold, Robert Kovacs, Min Deng, Antonius Nauman, Konrad Gerlach, **Horatio Hamkins**, Helena Lendowski, Chiao Fang, Shohei Katakura, Conrad Lemper, Muhammad Abdullah, and Patrick Baudisch (2026). AirForce: Personal Fabrication of Large-Scale, Load-Bearing Animatronics Structures from a Single Tube. *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)* [\[link\]](#)

Ergys Subashi, **Horatio Hamkins**, Samuel Hellman, Neelam Tyagi (2023). 3D-Printed Multi-Contrast Phantom for Quality Assurance of Adaptive Radiotherapy Delivery in a High-Field MR-Linac. *UCLA 9th MR in RT Symposium, Feb 2023* [\[link\]](#)

Research Experience

Medical Physics Research Intern

Summer 2024, Summer 2025

University of Texas MD Anderson Cancer Center — Houston, TX

- Developed software for medical physicists to automate radiation dosage delivery calculations using 3D printed phantoms. Research paper about findings is currently under review in *3D Printing in Medicine*.

Human Computer Interaction Research Intern

Jul 2023 – Dec 2023

Hasso-Plattner Institute, University of Potsdam — Potsdam, Germany

- Designed an easily manufactured check valve, reducing assembly time of large soft-robotics structures by 90%. Improved an Eulerian-path planning algorithm to minimize the number of additional hardware components needed.
- Assisted with editing of academic papers and accompanying video submissions.

Medical Physics Research Intern

Jun 2022 – Aug 2022

Memorial Sloan Kettering Cancer Center Research Institute — New York City, NY

- Developed 3D printed multimaterial life-size replicas to further research on quality assurance of radiotherapy.
- Invented a process that allows dosimeter and multimaterial insert placement without distorting the phantom's shape during radiotherapy treatment.

Machine Learning Intern

Jun 2020 – Aug 2020

University of Oxford, Computational Health Informatics Lab - Oxford, UK

- Worked with a team using PyTorch create CNNs for filtering out irregularities and background noise in low-cost electrocardiogram and photoplethysmography sensors that help hospital staff predict the severity of infectious diseases, including COVID-19, facilitating a better allocation of resources.

Industry Experience

Founder and Chair of Interdisciplinary Makerspace

August 2024 – June 2026

University of Bath — United Kingdom

- Established a Makerspace with 150 active users (including grad students and faculty) by securing funding, facilitating student engagement through workshops, and ensuring operational continuity through development of financial systems.

Software Engineer and CAD Drafter Intern

Jan 2024 – May 2024

New Project (Custom Art Fabrication Studio)— New York City, NY

- Created Rhino 3D scripts to speed up CAD design and CNC cutting workflows.
- Assisted with project management for a \$100-million-dollar overseas art installation.

Grants

- 2026 **Alumni Fund Grant For Makerspace (£2,000)**, University of Bath Alumni Association
- 2026 **Continuity of Makerspace Equipment (£1,500)**, Center For Excellence In Engineering & Design Education
- 2025 **Makerspace Equipment Grant (In-Kind ~£50,000)**, SETsquared Bath Innovation Center
- 2025 **Reimagining Spaces for Better Research Culture (£4,000)**, Research Culture England
- 2025 **Community Building: Expanding the Robot Fighting League**, Student Union, University of Bath
- 2025 **Mechanical Engineering Makerspace Innovation Grant**, University of Bath
- 2024 **Community Building: Creating a Robot Fighting League**, Student Union, University of Bath
- 2024 **Seed Funding For Makerspace 3D Printer Purchases**, University of Bath
- 2024 **Library Spark Competition Finalist**, University of Bath

Awards

- 2026 **Arthur Clements Award (£1500)**, University of Bath Alumni Association
Awarded to help fund a development of an e-ink tablet with an dynamic notebook-like user interface.
- 2026 **Shortlisted for Student Education Award**, University of Bath
For founding and leading an interdisciplinary student-run makerspace serving the entire University of Bath community.
- 2024 **Best Potential Clinical Impact Award**, MD Anderson Cancer Center
Awarded for research presentation, 'Automating 4D Patient-Specific Phantom Development for Adaptive Radiotherapy Quality Assurance.'

Projects

Portrait Drawing Robot

github.com/horatioh13/Portrait-Draw

- Created an autonomous drawing robot by developing a software pipeline that converts photos into G-code for a modified 3D printer. Used to draw 100+ portraits at the University of Bath Groups Fair.

AI-Marker: An Autonomous Agent for Marking Business Coursework

docs.google.com/ai-marker

- As part of a Business for Engineering class, designed a workflow incorporating nine separate AI agents to facilitate the autonomous marking of university coursework.

Carbon Fibre Brick Production Factory Group Project

- Designed a factory assembly line to produce 3,000 bricks per day, optimized for the lowest energy cost per brick.